MARS Robot Setup, Usage, and Development Notes

Project Overview

The MARS (Makerspace Autonomous Robot System) project was developed as a robotics platform for makerspace mobility and perception through the Makerspace Digital Twin initiative. The robot uses mecanum wheels for omnidirectional movement, a Jetson Orin Nano and Teensy 4.1 connected to an Xbox controller for control, and a ZED 2i camera with a custom YOLOv11 model to detect CNC machines and support future digital twin research.

The project focused on developing the foundational robotics hardware, embedded systems, and perception pipeline needed for future autonomous robotics and semantic mapping research within the Ingram Hall Makerspace environment.

Completed Features

- Omnidirectional mecanum wheel mobility
- Jetson Orin Nano to Teensy 4.1 motor communication
- Xbox controller remote-control operation
- PWM motor control implementation
- ZED 2i stereo camera integration
- YOLOv11 CNC machine detection proof-of-concept
- Human tracking and body detection using ZED SDK
- Rear LCD display integration
- Cylindrical aluminum chassis and acrylic dome assembly

Hardware Components

Main Compute and Embedded Systems

- NVIDIA Jetson Orin Nano

- Teensy 4.1 microcontroller
- Xbox controller
- Rear LCD display

Perception Systems

- ZED 2i stereo depth camera

Mobility Systems

- 4x mecanum wheels
- 4x DC gear motors with encoders
- Motor drivers

Power Systems

- 12V 100Ah LiFePO4 battery
- 12V to 5V DC-DC converter
- Power distribution wiring

Mechanical Systems

- Cylindrical aluminum chassis
- Acrylic dome enclosure
- Internal mounting structure

Power Distribution Notes

- The Jetson Orin Nano was connected directly to the battery power system.
- The 12V to 5V DC-DC converter was used to power lower-voltage electronics such as the Teensy 4.1, ZED 2i camera, and LCD display.

- Proper polarity and voltage verification should always be checked before powering the robot.
- Cable management and connector stability are important for safe operation and troubleshooting.

Control Pipeline

The robot control pipeline functions as follows:

Xbox Controller

→ Jetson Orin Nano

→ Control / command scripts

→ Teensy 4.1

→ PWM motor control

→ Motor drivers

→ DC motors

→ Mecanum wheel movement

Perception Pipeline

The computer vision and perception pipeline functions as follows:

ZED 2i Stereo Camera

→ Jetson Orin Nano

→ ZED SDK camera feed

→ YOLOv11 CNC detection model

→ Real-time CNC machine detections

→ Human tracking and body detection

→ Display output to monitor or LCD

YOLOv11 CNC Detection Model

The CNC detection model was developed as a proof-of-concept for identifying machines within the Ingram Hall Makerspace.

Dataset Information

- Dataset platform: Roboflow
- 4 CNC machine classes
- 95 original annotated images
- 229 total images after augmentation
- Trained using YOLOv11 Nano
- Used for real-time CNC machine detection

Augmentations Used

- Brightness adjustments
- Exposure adjustments
- Blur
- Noise
- Small image rotations

The trained model successfully demonstrated real-time detection of CNC machines using the ZED 2i camera feed.

Software Setup Notes

GitHub Repo link: [TXST-RAS/makerspace-detector: Runs Zed CV code to detect machines and human bodies in Ingram School of Engineering Makerspace]

Jetson Orin Nano

- Install the ZED SDK and required dependencies.
- Install Python packages required for YOLOv11 and perception scripts.
- Verify camera connectivity before running perception pipelines.
- Password: Ieeerobot!cs

Teensy 4.1

- Upload the motor control firmware to the Teensy.
- Verify PWM outputs and encoder feedback connections before testing motors.

Xbox Controller

- Connect or pair the Xbox controller with the Jetson system before starting control scripts.

YOLOv11 Model

- Place the exported YOLOv11 model file in the expected model directory.
- Verify the correct model path before launching detection scripts.

Robot Usage Steps

- Verify all wiring and power connections.
- Ensure the battery is connected safely and securely.
- Power on the Jetson Orin Nano and supporting electronics.
- Connect or pair the Xbox controller.
- Start the Teensy motor control firmware.
- Launch the Jetson control scripts.
- Slowly test mecanum wheel movement and steering.
- Start the ZED 2i camera pipeline.
- Launch the YOLOv11 detection model.

- Verify CNC machine detections and human tracking outputs.
- Shut down software safely before disconnecting power.

Known Limitations

- Autonomous navigation was not fully implemented.
- SLAM and Nav2 integration remain future work.
- CNC detection was implemented as a proof-of-concept using a limited dataset.
- Additional testing is recommended across more lighting conditions and machine viewpoints.
- Cable management and long-term power distribution can be further improved.
- Motors currently are small, but newer and larger motors (square-head faced) are available in locker to implement.

Future Development

Potential future work includes:

- Autonomous navigation using SLAM and Nav2
- Semantic mapping of the Makerspace
- Expansion of the CNC detection dataset
- Detection of additional makerspace equipment
- Improved cable management and enclosure organization
- Interactive user interface development
- Digital twin integration
- Autonomous workstation navigation
- Voice or touchscreen interaction systems

Key Takeaways

The project demonstrated the importance of integrating mechanical design, embedded systems, onboard power systems, and computer vision into a unified robotics platform. Real-world assembly and testing also showed that CAD designs often require iteration due to mounting constraints, wiring limitations, and physical tolerances not initially identified during design.